



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2024 – 0546



Agricultural Machinery – Heated-Air Mechanical Grain Dryer SHIN HEUNG NCD-120SBX

Test Report Number	2024 – 0546
Date of Issuance	December 3, 2024
Valid Until	December 3, 2029
Agricultural Machinery	Heated-Air Mechanical Grain Dryer
Type	Batch (Recirculating), Direct, Non-renewable, Tested for Rice
Brand	SHIN HEUNG
Model	NCD-120SBX
Test Requested by	BIO-CHEM TECHNOLOGY PHIL. CORP. Brgy. Makiling, Calamba City, Laguna; and SUNRIPE AGRI-MACHINERIES, INC. Unit 6, Mokara Bldg., Maharlika Highway, Brgy. Abar 1 st , San Jose City, Nueva Ecija

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 201:2015	AMTEC Test
Moisture Gradient, %, maximum	2.0 ^a	0.600
Drying Rate, %/h	0.8 ^a	1.19
Product Quality ^b		
a. Cracked Grain, % increase, maximum	3.0 ^a	10.00
b. Head rice, % decrease, maximum	5.0	14.65
c. Hulled/ Damaged Grain, % increase, maximum	2.0	0
Drying Loss/Spillage, %, maximum	1.0	0
Drying Efficiency, %, minimum	75	82.39
Heating System Efficiency, %, minimum	90 ^c	91.86
Blower flow rate, cmm/kg (cmf/ton)	0.05 (1750)	NM ^d
Static Pressure, mm H ₂ O (in H ₂ O)	25 (1)	NA
Grain Holding Capacity, kg	12000 ^e	12062
Discoloration, Traces of Unburned Fuel, Ashes on Grain Surface	Shall not be present	None
Fermented or Musty Smell	Shall not be present	None
Thermometer	Shall be provided	Provided
Pressure Gauge	Shall be provided	Provided
Noise Level, dB(A), maximum	92	81.7

^a Requirement for batch type recirculating heated-air mechanical grain dryer.

^b Allowable difference between the laboratory analysis and machine's performance.

^c Requirement for direct-fired petroleum-based fuel.

^d The weight of the grains in the drying chamber cannot be accurately measured at a given point during drying.

^e As specified by the manufacturer.

Note: NA – Not Applicable (The dryer is diesel-fed recirculating type.), NM – Not Measured

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	2022 – 0130, 2022 – 0454, and 2023 – 0265
3. Manufacturer	BIO-CHEM TECHNOLOGY PHIL. CORP.
4. Serial number	120SD-24-201004
5. Heat source	SH G15BX Diesel-fed burner
6. Safety controls/devices	The dryer has a self-diagnostic code and an automatic alarm to indicate abnormalities.
7. Dust collection systems	The dryer is equipped with a dust collection system consisting of blowers and a cyclone.
8. Instrumentation, setting and adjustments	The dryer is equipped with a digital main control panel for setting, regulating, monitoring, and controlling the drying operation. It includes digital displays for drying time, drying air temperature, grain temperature, ambient temperature, and grain moisture content. The drying time, drying temperature and grain temperature can be set to automatically control the heat produced from furnace by controlling the flow of biomass automatically using the auger and blower located below at the biomass input hopper. The flow of air can be manually adjusted using regulating gates on the furnace.

OBSERVATIONS (Cont'n)

9. Loading and unloading	The test material is loaded manually through the input loading hopper of the pre-cleaner. The dried product is discharged through the outlet pipe at the top of the elevator through a directional valve going either back to the drying bin or to the bagging bin and then bagged manually using sacks.
10. Number of operators needed to load the grains	Five (Two loaders and three haulers)
11. Number of operators needed to load the biomass (rice hull)	Not applicable
12. Shall have adequate provision for fire control	The dryer has a thermal sensor for abnormal temperature detection.
13. Shall have adequate protection from or for all moving parts	Some transmission systems are provided with cover.
14. Shall be provided with features for access to parts during repair, maintenance, and operation	Parts can be accessed for repair and maintenance through the removable panels of the machine and ladder.
15. The noise emitted by the heated-air mechanical dryer shall not exceed 92 dB(A).	The sound level emitted by the heated-air mechanical dryer during operation was 81.7 dB(A) taken at the feeding operator's ear level.
16. Provision shall be made for dust control. The dust emission of the machine shall be within the acceptable limits set by the Department of Environment and Natural Resources.	The machine is equipped with a dust collector system at the end of the blower, and ash cyclone and ash collection bins for proper ash disposal.
17. Shall be free from defects that may be detrimental to its use	The dryer is free from defects that may be detrimental to its use.
18. Shall be free from sharp edges and surfaces that may hurt the operator	The dryer is free from sharp edges and surfaces.
19. All metal parts should be machine bent, pressed and cut to avoid rough surfaces. All rough surfaces should be machine finished and smoothed.	No rough surfaces were observed on the components of the machine.
20. The warning notice shall be in accordance with PNS/BAFS 330:2022 – Agricultural Machinery – Technical Means for Ensuring Safety – Guidelines.	Warning notices are provided on the machine (Figure 7 and 8).
21. Provision shall be made so that it can withstand extreme weather conditions to which it can be exposed.	The dryer is installed inside a permanent structure that could withstand extreme weather conditions.
22. Other remarks	The heated-air mechanical grain dryer can also be used with a biomass-fired furnace.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	November 6-7, 2024
Location of Test	Brgy. Caridad Sur, Llanera, Nueva Ecija 15°44'07.73" N, 121°01'01.61" E
Items Tested	Operator's Manual, Drying Capacity, Moisture Reduction Rate, Heating System Efficiency, Drying Efficiency, Drying System Efficiency, Moisture Content Variation, Moisture Content Gradient, Fuel Consumption, Power Consumption, Number of Operators, and Quality of Dried Grains

METHODS OF TEST

The grain dryer was tested based on the PNS/PAES 202:2015 Agricultural Machinery – Heated-air Mechanical Grain Dryer – Methods of Test.

DESCRIPTION OF THE MACHINE

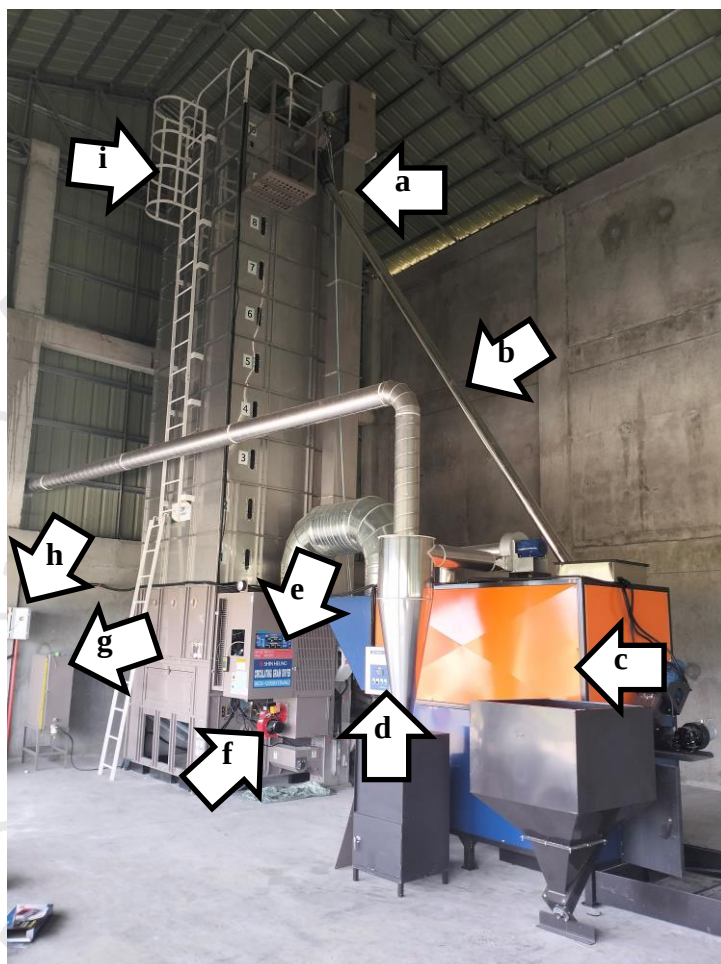


Figure 1. Different parts of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer and its components heated by SH G15BX Diesel-fed burner include the following:
(a) Elevator, (b) Output chute, (c) Biomass furnace, (d) Furnace control panel,
(e) Dryer control panel (f) Diesel-fed burner, (g) Fuel tank,
(h) Circuit breaker, and (i) Access ladder.

DESCRIPTION OF THE MACHINE (Cont'n)



Figure 2. Different parts of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer and its components heated by SH G15BX Diesel-fed burner include the following: (a) Drying chamber, (b) Input loading hopper, (c) Plenum, (d) Pre-cleaner, and (e) Built-in moisture meter.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Drying rate, %/h	>0.8	1.19 ^b
2 Overall dimensions, mm (installed)		
2.1 Length	3460/8415	8350 ^c
2.2 Width	2600	4750
2.3 Height	9387	9270
3 Drying bin		
3.1 Type	Vertical	Vertical, rectangular
3.2 Dimensions, mm		
3.2.1 Length	2000	2040
3.2.2 Width	1870	1850
3.2.3 Height	8410	855
3.3 Holding capacity, kg	12000	12062 ^b
3.4 Maximum grain depth, mm	NA	NM ^d
3.5 Material		
3.5.1 Frame	Stainless steel	304SS/303SS ^d
3.5.2 Holding bin	Stainless steel	304SS/303SS ^d
3.5.3 Wall	Stainless steel	304SS/303SS ^d
4 Fan		
4.1 Type	Centrifugal	Axial
4.2 Brand and model	SH	SH ^a
4.3 Air flow rate, m ³ /min	220	222.38
4.4 Static pressure, Pa	40	NA ²
4.5 Material	Steel	ND ^e
4.6 Prime mover	2.2 kW Electric Motor	2.2 kW Electric Motor ^a
5 Heater		
5.1 Type	Gun Type Burner, Diesel (Direct fired)	Diesel-fed burner
5.2 Brand and model	SH G15BX	SH G15BX ^a
5.3 Fuel	Diesel	Diesel
5.4 Heat output, kJ/h	490000	482703.51 ^b
5.5 Fuel consumption rate, L/h	12	13.60 ^b

^a The data were obtained from the specifications sheet provided by the test applicant.

^b Based on actual test results.

^c Biomass furnace was included.

^d The depth of grains in the drying bin cannot be accurately measured at a given point during drying.

^e Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer; see Table 1 for the elemental composition of the material.

Note: NA – Not Applicable (The dryer is diesel-fed recirculating type), ND – No Data, NM – Not measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification	AMTEC Verification
5.6	Capacity of fuel tank, L	200	202
5.7	Mechanism for fuel feed	Geared pump	Fuel pump
5.8	Method of temperature control	Automatic control	Automatic control
5.9	Material	Burner case casting and chamber castable	Metal sheet
6	Material handling system		
6.1	Input loading hopper		
6.1.1	Dimension, mm		
6.1.1.1	Length	1230	875
6.1.1.2	Width	ND	380
6.1.1.3	Height	440	135
6.1.2	Material	Electronic Galvanized Iron	304SS/303SS ^b
6.1.3	Other features	ND	Equipped with pre-cleaner
6.2	Elevators		
6.2.1	Number	1	1
6.2.2	Height, m	9000	9030
6.2.3	Capacity, t/h	12	12 ^a
6.2.4	Belt dimensions, W × t, mm	215	215 ^a
6.2.5	Elevator buckets		
6.2.5.1	Dimensions, L × W × t, mm	210 × 81 × ND	210 × 81.28 × 5.42
6.2.5.2	Material	Nylon66	Hard plastic
6.2.6	Drive motor		
6.2.6.1	Type	Induction motor	Induction motor
6.2.6.2	Rated power, kW	1.5	1.5 ^a
6.2.6.3	Rated speed, rpm	1730	1730 ^a
6.2.6.4	Electric service required	Three-phase	Three-phase
6.2.6.5	Current, A	6.8	6.8 ^a
6.2.6.6	Voltage, V	220	220 ^a
6.2.6.7	Frequency, Hz	60	60 ^a
6.2.7	Material	Stainless steel	304SS/303SS ^b
6.2.8	Other features	Microprocessor and program (automatic)	None

^a The data were obtained from the specifications sheet provided by the test applicant.

^b Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer; see Table 1 for the elemental composition of the material.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification	AMTEC Verification	
7	Instruments and controls		
7.1	Drying air temperature	Sensor	Drying air temperature sensor
7.1.1	Location	ND	Inside the plenum
7.2	Grain temperature	Sensor	Thermocouple
7.2.1	Location	ND	Between the grain conveyor and elevator
7.3	Grain moisture content	Integral automatic single kernel moisture meter	Integral moisture meter with controller
7.3.1	Location	Bucket elevator	Outside the casing of the elevator.
8	Safety features		
8.1	Product safety	Automatic temperature control for drying air and grain temperature, moisture meter and abnormalities indicator	Automatic temperature control, abnormal indicator/alarm, and automatic grain moisture meter.
8.2	Machine safety	Provided with alarm, overheat protector, shutter drum limit switch, automatic stop if malfunction, flame detector, safety relay, overload detector, airflow switch, thermal sensor.	Automatically stops drying when the machine experiences malfunctions.
9	Operator safety	Belt cover, loading door protection net, railings, dust control, flame detector and warning stickers	Provided with warning notices on the controller, some belt cover for the transmission system, dust collection system, and railing on the top of the dryer.

^a The data were obtained from the specifications sheet provided by the test applicant.

Note: ND – No Data

RESULTS

Item		Data
1	Performance test	
1.1	Initial weight of test material, kg	12062
1.2	Final weight of test material, kg	9989
1.3	Initial moisture content, % _{wb}	28.53
1.4	Final moisture content, % _{wb}	13.70
1.5	Actual drying time, h	12.5
1.6	Drying capacity, kg/h	964.96
1.7	Rate of moisture removal, kg/h	165.86
1.8	Moisture reduction rate, %/h	1.19
1.9	Average drying air temperature, °C	61.2
1.10	Ambient air condition	
1.10.1	Average dry bulb temperature, °C	26.9
1.10.2	Average wet bulb temperature, °C	25.4
1.10.3	Relative humidity, %	88.4
1.11	Exhaust air condition	
1.11.1	Average dry bulb temperature, °C	35.0
1.11.2	Average wet bulb temperature, °C	31.7
1.11.3	Relative humidity, %	79.7
1.12	Average grain temperature, °C	35.2
1.13	Fan air velocity, m/s	6.30
1.14	Total air flow rate, m ³ /min	222.38
1.15	Plenum static pressure, mmH ₂ O	NA
1.16	Fuel consumption, L/h	3.79
1.17	Power consumption	13.60
1.17.1	Total electrical input power, kW	220.0
1.17.2	Line voltage, ACV	81.7
1.17.3	Load current, A	79.0
1.18	Noise level at operator's ear level, dB(A)	
1.18.1	Feeding operator	81.7
1.18.2	Machine operator	79.0
1.19	Heat utilization, kJ/kg, water removed	2910.33
1.20	Heat system efficiency, %	91.86
1.21	Drying efficiency, %	82.39
1.22	Drying system efficiency, %	75.68

Note: NA – Not Applicable (The dryer is diesel-fed recirculating type)

RESULTS (Cont'n)

Item		Data	
1.23	Moisture content variation at the end of drying, %	0.200	
1.24	Moisture content gradient, %	0.600	
2	Condition of test material	Control	Dryer Test
2.1	Variety	R222	
2.2	Source	Llanera, Nueva Ecija	
2.3	Paddy analysis		
2.3.1	Grain size, brown rice, mm		
2.3.1.1	Length	7.10	7.47
2.3.1.2	Width	2.11	2.17
2.3.1.3	Thickness	1.48	1.51
2.3.2	Grain type	Long and slender	Long and slender
2.3.3	Paddy bulk density, kg/m ³	466.50	505.19
2.3.4	Moisture content, % _{wb}	13.90	13.70
2.3.5	Purity, %	96.71	98.28
2.3.6	Cracked, %	1.67	11.67
2.3.7	Immature, %	1.28	1.84
2.3.8	Chalky, %	0.213	0.187
2.3.9	Yellow and fermented, %	6.32	6.95
2.3.10	Red rice, %	0.040	0.400
3	Laboratory milling test ^a		
3.1	Total milling recovery, %	62.03	61.24
3.2	Head rice recovery, %		
3.2.1	Based on milled rice	87.89	73.24
3.2.2	Based on rough rice	54.52	44.85
3.3	Broken rice, %		
3.3.1	Based on milled rice	10.25	23.46
3.3.2	Based on rough rice	6.36	14.37
3.4	Brewer's rice, %		
3.4.1	Based on milled rice	1.86	3.30
3.4.2	Based on rough rice	1.15	2.02

^a Satake laboratory test

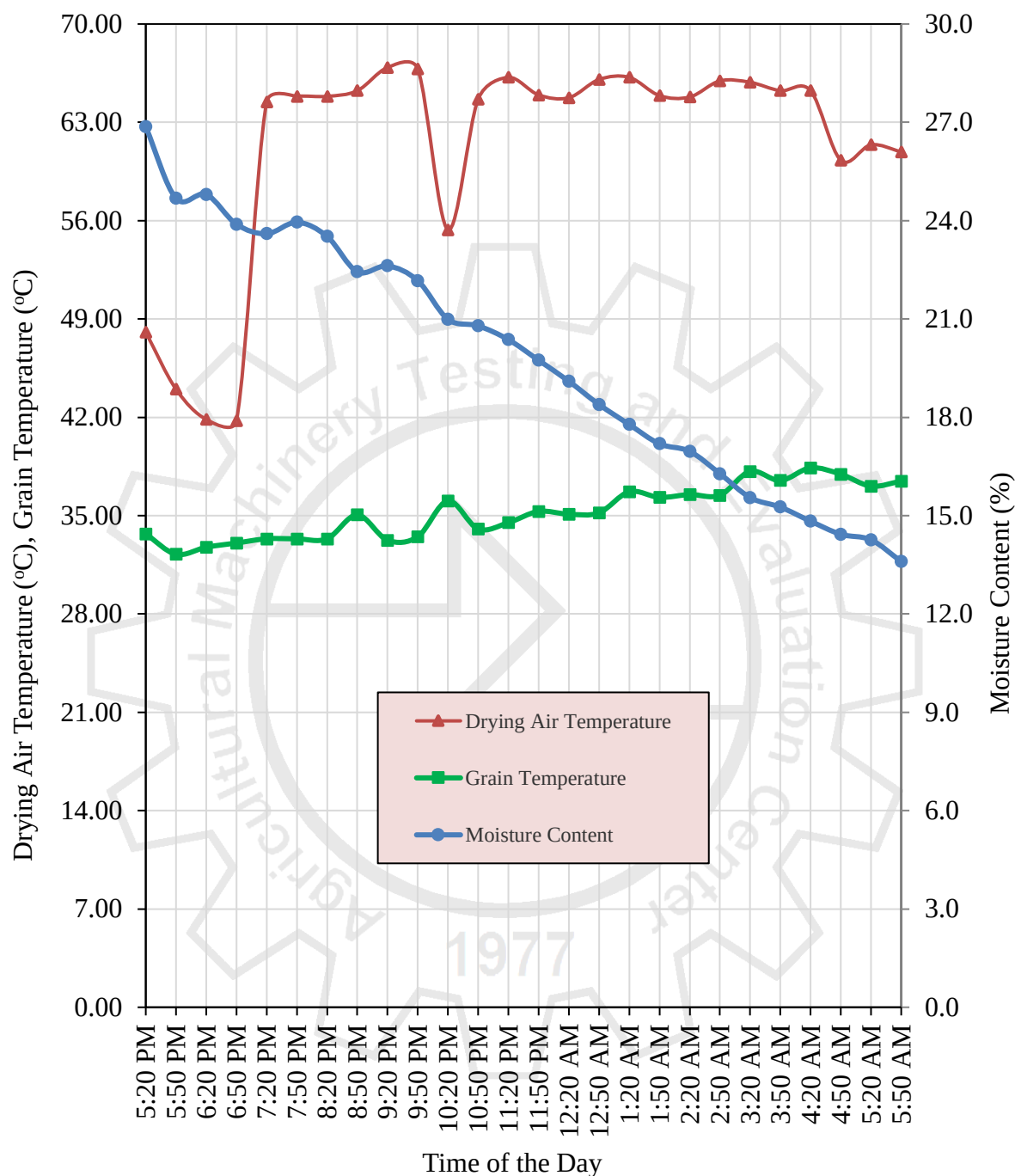
RESULTS (Cont'n)

Figure 3. Drying curve of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer

RESULTS (Cont'n)

Table 1. Elemental composition of the different components of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

Element	Percent Composition ^a		
	Pre-cleaner hopper and body, and dryer hopper and elevator body	Pre-cleaner hopper and body, and dryer hopper and elevator body	Grain conveyor
Aluminum	<LOD	<LOD	<LOD
Silicon	0.4412	0.4412	<LOD
Phosphorus	0.0250	0.0250	0.7135
Sulfur	<LOD	<LOD	0.0251
Titanium	<LOD	<LOD	<LOD
Vanadium	0.1264	0.1264	<LOD
Chromium	18.5282	18.5282	0.0438
Manganese	1.0909	1.0909	18.8159
Iron	71.1993	71.1993	1.0876
Cobalt	0.3535	0.3535	70.9430
Nickel	8.0721	8.0721	<LOD
Copper	0.0236	0.0236	7.9259
Zirconium	<LOD	<LOD	0.1618
Niobium	<LOD	<LOD	<LOD
Molybdenum	0.0276	0.0276	0.0089
Tin	0.0471	0.0471	0.1232
Tantalum	0.0143	0.0143	0.0958
Tungsten	<LOD	<LOD	<LOD
Lead	<LOD	<LOD	<LOD
Grade	304SS/303SS	304SS/303SS	304SS/303SS

^a Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS

Figure 4. Nameplate of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

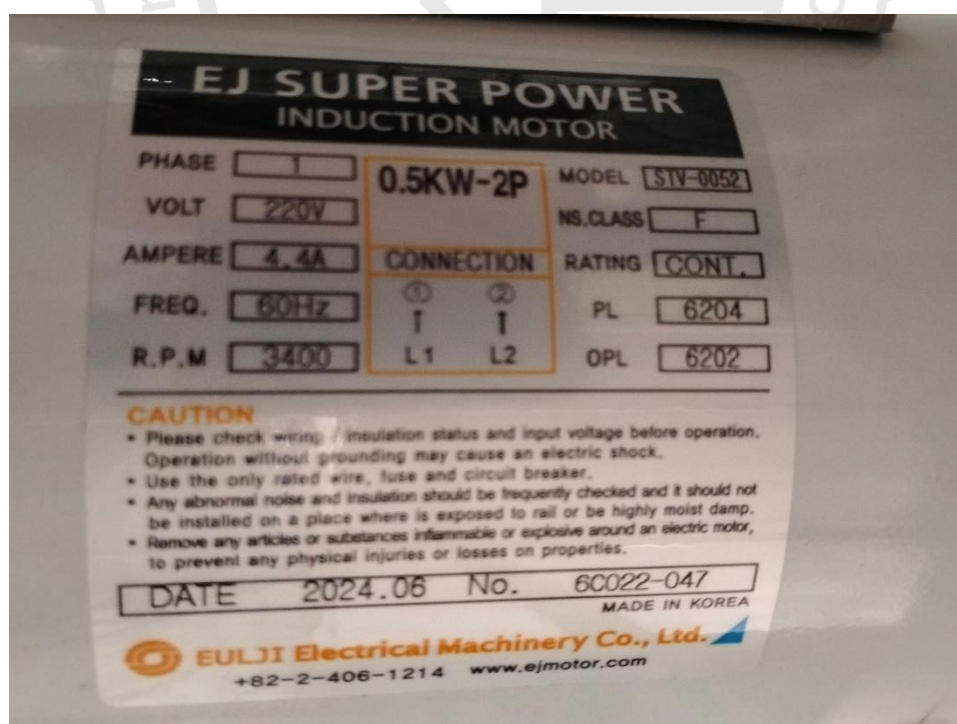


Figure 5. Nameplate of the dust collection system's motor of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

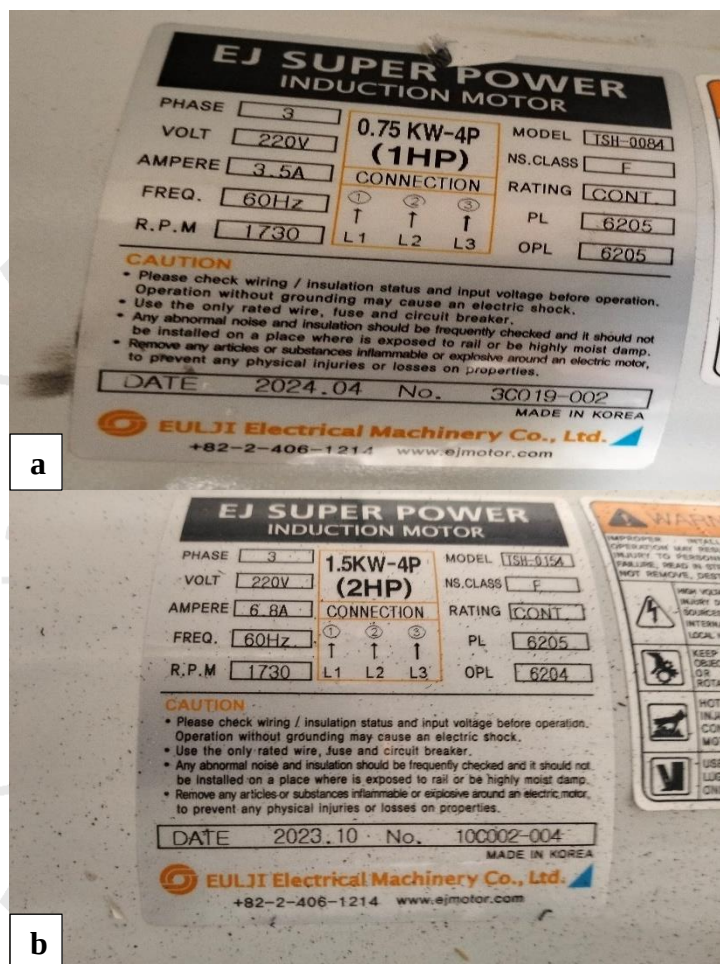
OBSERVATIONS (Cont'n)

Figure 6. Nameplate of the (a) pre-cleaner motor and (b) elevator motor of the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

OBSERVATIONS (Cont'n)

Figure 7. Markings and safety warnings on the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

OBSERVATIONS (Cont'n)

Figure 8. Safety warnings on the SHIN HEUNG NCD-120SBX Heated-Air Mechanical Grain Dryer.

Tested by:


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Approved for Release:


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AMTEC Director

11 DEC 2024
Date

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